Causal Machine Learning

HHU summer term 2026

April 15, 2026

Prof. Dr. Jannis Kück

About me

- Professor of Data Science in Economics
- Mainly working on development of new statistical/ machine learning methods for the estimation of causal parameters in econometric models
- Research Interest: High-dimensional Statistics, Causality, Machine Learning
- Teaching: Causal Inference, Machine Learning, Econometrics, Statistics, Programming
- Contact: kueck@dice.hhu.de

Organizational Matters

Lectures

- Wednesday (16:30-18:00h)

Tutorials

- Thursday (08:30 - 10:00h)
- We will work with R Notebooks. You can use R in VS Code or Google Colab for this.
- The notebooks will be provided here.

Grading

- Oral exam at the end of the semester

Organizational Matters

Lectures

- Wednesday (16:30-18:00h) Please note that there will be no lecture on July, 1!

Tutorials

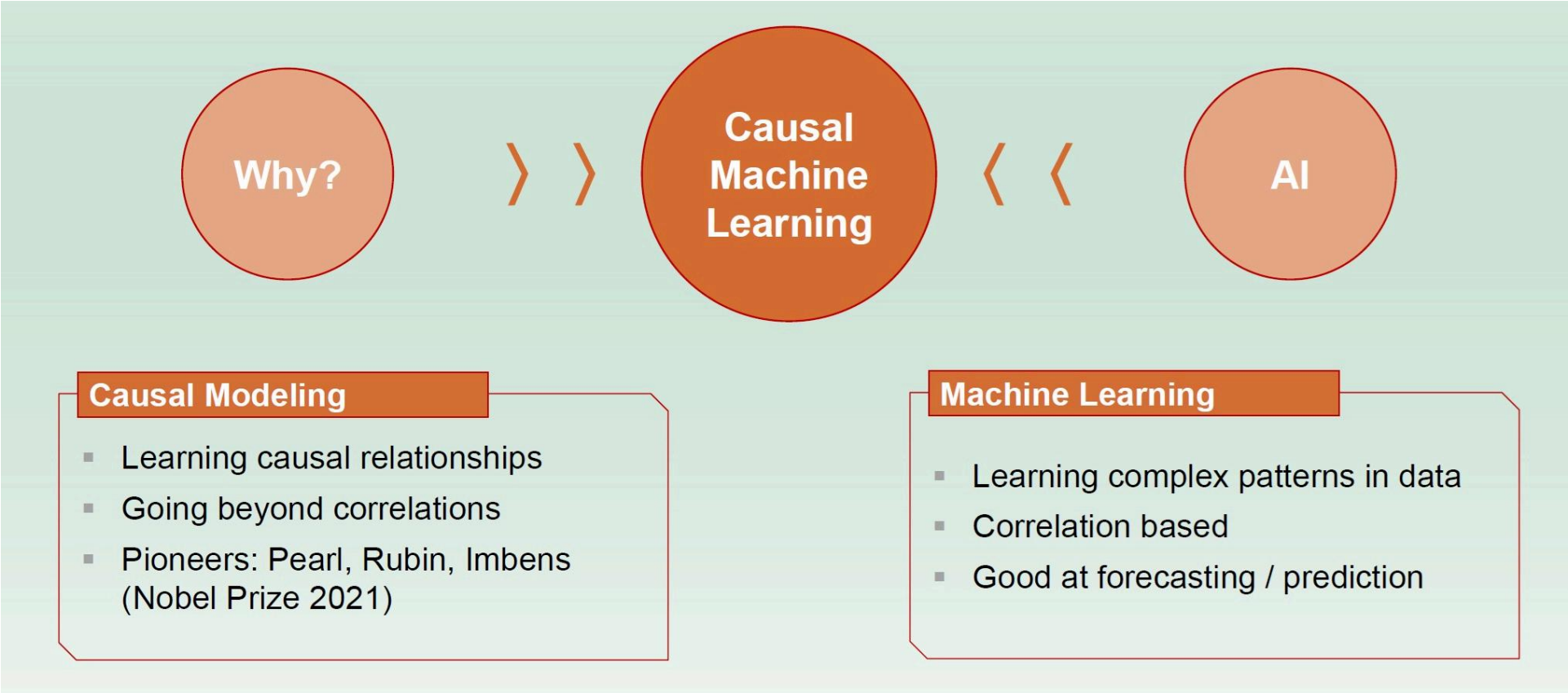
- Thursday (08:30 - 10:00h) Please note that there will be no tutorial on July, 2!

Course Outline

The course will cover the following topics:[1]

- Predictive Inference with Linear Regression
- Basics in Causal Inference (Randomized Experiments, ...)
- Machine Learning Methods (Lasso, Random Forests, ...)
- Statistical Inference in High-dimensional Linear Regression
- Causal Inference via Conditional Exogeneity
- Structural Equations and DAGs
- Statistical Inference in Modern Nonlinear Regression
- Double Machine Learning
- Optional Advanced Topics (high-dimensional IV models, Difference-in-Differences with Machine Learning, Panel data, Sensitivity Analysis, ...)

Causal machine learning



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Prediction vs. Causal Inference

Predictive ML	Causal ML
How can we build a good prediction rule, $f(X)$, that uses features X to predict? **Example:** Customer Churn *"How well can we predict whether customers churn?"*	What is the causal effect of a treatment D on an outcome Y? **Example:** Customer Retention *"Why do customer churn?"* *"How can we retain customers?"*

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Causal Inference

Causal Modeling

- Look into the black box and **learn about the underlying mechanism**
- Example: **Association** between lung cancer and wearing a lighter
  - Obviously, **no causal relationship**
  - Nevertheless, prediction might work!
- **Causal understanding is key for real "Intelligence"**

Effectiveness of Marketing Expenses

Resource Allocation

Customer Churn

Returns in Retail

Pricing

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Causal Inference and Machine Learning

Causal Inference	ML

• Experminents vs. observational data
• Tools for learning from observational data:
  • Uncoundedness assumption
  • Instrumental Variables
  • Regression Discontinuity
  • Difference-in-difference
  • …
• Not covered: causal discovery
• Heterogeneity has important implications.
	• Supervised vs. unsupervised learning
	• Supervised learning algorithms:
	  • Lasso / Ridge / Elastic Net
	  • Regression Trees
	  • Random Forest and Boosted Trees
	  • Neural Nets
	• Designed for prediction in high-dimensional settings

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What to expect

- Recap/learn the **Causal Inference** and **Machine Learning** basics
- Learn econometric models for high-dimensional data
- Learn recent Causal ML methods (e.g. **Double Machine Learning**) for estimation of (average) treatment effects
- Get to know recent software (e.g. DoubleML package in R) to apply Causal ML algorithm in practice
- Guest lecture with one of the developers of the DoubleML package

Some suggested textbooks

- Trevor Hastie, Robert Tibshirani & Jerome Friedman: The Elements of Statistical Learning
- Scott Cunningham: Causal Inference: The Mixtape
- Miguel A. Hernan and James M. Robins: Causal Inference, What If
- Judea Pearl, Madelyn Glymour, Nicholas P. Jewell: Causal Inference in Statistics: A primer
- Victor Chernozhukov, Christian Hansen, Nathan Kallus, Martin Spindler and Vasilis Syrgkanis: Applied Causal Inference powered by ML and AI
- Martin Huber: Causal Analysis: Impact Evaluation and Causal Machine Learning with Applications in R
- Great causalML course by Michael Knaus (University of Tübingen) with a focus on impact/policy/program evaluation.

Credits

The course is based on the book “Applied Causal Inference powered by ML and AI” (available here). I would like to thank the authors Victor Chernozhukov, Christian Hansen, Nathan Kallus, Martin Spindler and Vasilis Syrgkanis for sharing the materials.

Causal ML